

PKU-Zurich Summer School

Lecture-3: Deep Learning for Macro-Finance with Household Heterogeneity

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Outline

Motivation

Model

Richer Model

Methodology

Motivation

Motivation

We saw how to solve PDEs using deep learning in the last lecture. This has been used to solve a variety of models in macro-finance.

- Examples include Fernandez-Villaverde et al (2024), Duarte et al (2024), Gu et al. (2025), and Gopalakrishna and Wu (2026).
- In principle, these methods should make it possible to solve models that were previously computationally intractable.
- In practice, researchers find it difficult to use these methods for problems with wealth distributions, long-lived assets, and complicated portfolio choice.

In this lecture, we discuss how to:

1. characterize equilibrium in heterogeneous-agent macro-finance models as a collection of PDEs,
2. train neural networks to solve the PDEs.

Motivation

We consider a class of DSGE models with the following features:

1. Goods are produced by a Lucas tree according to a stochastic process.
2. There is a short-term risk-free asset and a set of risky assets with claims to the tree.
3. There is a large collection of price-taking agents who face uninsurable idiosyncratic shocks or portfolio constraints.

Three blocks

General equilibrium for this economy can be characterized by three blocks

1. **Optimization block:** High but finite dimensional PDE capturing agents optimization over consumption, portfolio holdings taking prices as given
2. **Distribution block:** Infinite dimensional low of motion of agent's wealth
3. **Equilibrium block:** Ensures that price processes are consistent with equilibrium

1. Optimization

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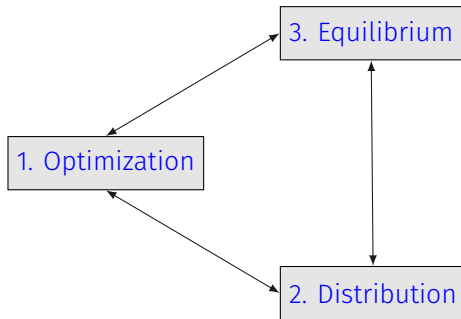
3. Equilibrium

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Model

Model Setup

Time is continuous with infinite horizon.

- There is one type of perishable consumption good created by a Lucas tree:

$$dy_t = \mu^y(y_t) dt + \sigma^y(y_t) dW_t,$$

where W_t is a standard Brownian motion.

- The economy is populated by a large collection of price-taking households with flow utility $u(c_{i,t})$ and wealth $a_{i,t}$.
- There are competitive markets for goods, risk-free bonds in net zero supply, and equity shares in the Lucas tree.
- The price process follows $dq_t = \mu_{q,t} q_t dt + \sigma_{q,t} q_t dW_t$
- Financial frictions restrict agents' consumption and portfolio choice:

$$\Psi(a_{i,t}, b_{i,t}, \kappa_i) \geq 0,$$

where $b_{i,t}$ is the amount of bond holdings, and κ_i is the penalty for holding capital.

Optimization

Given beliefs about price processes $\hat{r}$ and $\hat{q}$, agent i solves

$$\max_{c,b} \quad \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} (u(c_{i,t}) + \psi(a_{i,t}, b_{i,t}, \kappa_i)) dt \right]$$

subject to

$$\begin{aligned} da_{i,t} = & \left[b_{i,t} r_t - c_{i,t} + \left(\mu_t^q + \frac{y_t}{q_t} \right) (a_{i,t} - b_{i,t}) \right] dt \\ & + (a_{i,t} - b_{i,t}) \sigma_t^q dW_t. \end{aligned}$$

We derive two recursive characterizations:

1. One in terms of natural state variables:

$$(y, \{a_i\}_{i=1}^I).$$

2. One in terms of wealth shares:

$$(y, \{\eta_i\}_{i=1}^I), \quad \eta_i := \frac{a_i}{\sum_j a_j}.$$

HJBE

A finite-agent approximation gives an HJBE of the form

$$\begin{aligned} \rho V_i(a_i, \cdot) = \max_{c_i, b_i} & \left\{ u(c_i) + \psi(a_i, s_i, \kappa_i) + \frac{\partial V_i}{\partial a_i} \mu_{a_i}(c_i, b_i, \cdot) + \frac{\partial V_i}{\partial y} \mu_y \right. \\ & + \frac{1}{2} \frac{\partial^2 V_i}{\partial a_i^2} \sigma_{a_i}^2(b_i, \cdot) + \frac{1}{2} \frac{\partial^2 V_i}{\partial y^2} \sigma_y^2 + \frac{\partial^2 V_i}{\partial a_i \partial y} \sigma_{a_i}(b_i, \cdot) \sigma_y \\ & \sum_{j \neq i} \frac{\partial^2 V_i}{\partial a_i \partial a_j} \sigma_{a_j}(b_j, \cdot) \sigma_{a_j}(\cdot) + \sum_{j \neq i} \frac{\partial^2 V_i}{\partial a_j} \mu_{a_j}(\cdot) + \sum_{j \neq i} \frac{\partial^2 V_i}{\partial a_j \partial z} \sigma_{a_j}(\cdot) \sigma_z \\ & \left. + \frac{1}{2} \sum_{j \neq i, j' \neq i} \frac{\partial^2 V_i}{\partial a_j \partial a_{j'}} \sigma_{a_j}(b_j, \cdot) \sigma_{a_{j'}}(b_{j'}, \cdot) \right\}. \end{aligned}$$

Blue part: Effect of other agents' wealth dynamics on agent i 's value.

FOCs

Define the marginal value of wealth:

$$\xi_i := \frac{\partial V_i}{\partial a_i}.$$

The first-order conditions imply

$$\begin{aligned}\xi_i &= u'(c_i), \\ \xi_i(r - r^q) &= \sigma_{\xi_i} \sigma_q + \frac{\partial \xi_i}{\partial b_i}.\end{aligned}$$

The asset-pricing condition holds regardless of the state space that ξ_i lives on.

FOCs in Wealth Share Space

Rewriting the asset-pricing condition on the state space $(y, \{\eta_i\}_{i=1}^I)$ gives

$$\xi_i(r - r^q) = \left(\sum_j \frac{\partial \xi_i}{\partial \eta_j} \mu_{\eta_j} + \frac{\partial \xi_i}{\partial y} \mu_y \right) \sigma_q + \frac{\partial \xi_i}{\partial b_i}.$$

This allows us to work in wealth-share space going forward.

Block 1: Optimization

Given prices $(r, r^q, q, \mu^q, \sigma^q)$, agent optimization implies

$$0 = -\rho + r + \mu_{\xi_i} + \frac{1}{\xi_i} \frac{\partial \psi_i}{\partial a_i},$$

$$\xi_i = u'(c_i),$$

$$r - r^q = \sigma_{\xi_i} \sigma_q + \frac{\partial \psi_i}{\partial b_i}.$$

This block determines consumption, portfolio choice, and the drift and volatility of marginal value of wealth $(\mu_{\xi_i}, \sigma_{\xi_i})$.

Block 2: Distribution

Given prices $(r, r^q, q, \mu^q, \sigma^q)$ and policies (c, ξ, b) , the law of motion of wealth shares is

$$\begin{aligned}\mu_t^{\eta_j} &= \frac{b_{j,t}}{a_{j,t}} r_t - \frac{(u')^{-1}(\xi_{j,t})}{a_{j,t}} + (r_t^q - r_t) \frac{b_{j,t}}{a_{j,t}} - \frac{b_{j,t}}{\eta_{j,t} q_t} y_t + \left(\frac{b_{j,t}}{a_{j,t}} \sigma_t^q \right)^2, \\ \sigma_t^{\eta_j} &= \frac{1}{a_{j,t}} (a_{j,t} - b_{j,t}) \sigma_t^q - \sigma_t^q = -\frac{b_{j,t}}{a_{j,t}} \sigma_t^q.\end{aligned}$$

Block 3: Equilibrium

In equilibrium, the following market clearing conditions must hold:

$$\sum_i b_{i,t} = 0, \quad \sum_i \eta_{i,t} = 1, \quad \sum_i c_{i,t} = y_t.$$

The consistency conditions are

$$q\mu_t^q = \sum_j \frac{\partial q}{\partial \eta_j} \eta_j \mu_t^{\eta_j} + \frac{\partial q}{\partial y} \mu_{y,t} + \text{Covariance},$$

$$q\sigma_t^q = \sum_j \frac{\partial q}{\partial \eta_j} \eta_j \sigma_t^{\eta_j} + \frac{\partial q}{\partial y} \sigma_{y,t}.$$

Comparison to Other Methods

Model / Method	Optimization	Distribution	Equilibrium
RA, Lucas-style	trivial	NA	trivial
HA, Krusell–Smith	trivial	yes	trivial
Long-lived assets	trivial	NA	yes
HA + long-lived assets	yes	yes	yes

This lecture targets the case where all three blocks are non-trivial.

Richer Model

Environment: Setting, Production, and Households

- Continuous time $t \in [0, \infty)$. One perishable consumption good, one capital stock.
- Goods production technology $y_t = e^{z_t} k_t$, where capital $dk_t = (\Phi(l_t) - \delta)dt$
 - ▶ Aggregate productivity follows $dz_t = \alpha_z(\bar{z} - z_t)dt + \sigma_z dW_{z,t}$.

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- Goods production technology $y_t = e^{z_t} k_t$, where capital $dk_t = (\Phi(\iota_t) - \delta)dt$
 - ▶ Aggregate productivity follows $dz_t = \alpha_z(\bar{z} - z_t)dt + \sigma_z dW_{z,t}$.
- **Households:** Continuum of price-taking agents ($i \in [0, 1]$)
 - ▶ Idiosyncratic **retirement shocks** at rate λ_h : retired agents replaced with new wealth.
 - ▶ $u(c_{i,t}) = \beta c_{i,t}^{1-\gamma} / (1-\gamma)$ while young,
 - ▶ $u(c_{i,t}) = (1-\beta) c_{i,t}^{1-\Gamma} / (1-\Gamma)$ at retirement. [Microfounded]

Environment: Frictions on Household Portfolio Choices

- *Friction-1*: households cannot provide retirement insurance contracts to each other.
- *Friction-2*: penalty on holding assets $\psi_h^k(k_i, a_i)$, $\uparrow$ in k_i and $\downarrow$ in a_i
 - ▶ Will be disciplined to match household portfolio choices.
 - ▶ Interpretation: cost of participation in equity markets (e.g., from knowledge acquisition).
 - ▶ Interpretation: generalized non-homothetic preferences (e.g., *Carroll (1998)*).

$\Rightarrow$ non-degenerate density across household wealth, $g_h(a)$ and need for banks/funds:

$$dg_h(\omega) = \mu^g(\omega, \mathbf{S})dt + \sigma^g(\omega, \mathbf{S})dW$$

Environment: Financial Intermediaries, Govt., & Markets

- Two types of financial intermediaries:
 - ▶ **Bankers (b)**: issue risk-free deposits (at r_t^d) and hold assets.
 - ▶ **Fund managers (f)**: issue pension/insurance products and hold assets.
...a contract pays \$1 at retirement, priced q_t^n today.
 - ▶ $j \in \{b, f\}$ exit at rate $\lambda_{f,b}$ and pay dividends, and are replaced with new entrants.
 - ▶ **Friction**: They cannot raise outside equity.

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 - ▶ **Friction**: They cannot raise outside equity.
- **Government**: Issues zero-coupon bonds $M_t = \mathcal{M}K_t$ that mature at rate λ_m .
 - ▶ Raises wealth tax τ on living agents to finance debt.
- Competitive markets: Asset prices for capital, contracts, and bonds, $\mathbf{q}_t = (q_t^k, q_t^n, q_t^m)$.

Model: Optimization

- Household i with wealth $a_{i,t} := q_t^k k_{i,t} + q_t^n n_{i,t} + d_{i,t}$ solves

$$\begin{aligned} & \max_{c_i, \mathcal{C}_i} \mathbb{E}_0 \left[\int_0^T e^{-\rho t} (u(c_{i,t}) - \psi(k_{i,t}, a_{i,t})) dt + e^{-\rho T} \mathcal{U}(\mathcal{C}_{h,T}) \right] \\ & s.t. \frac{da_{i,t}}{a_{i,t}} = \theta_{i,t}^k dR_t^k + \theta_{i,t}^n dR_t^n + (1 - \theta_{i,t}^k - \theta_{i,t}^n) r_t^d dt - c_{i,t}/a_{i,t} dt - \tau_{i,t} dt \\ & \mathcal{C}_{i,T} \leq (1 - \theta_{i,t}^n + \theta_{i,t}^n/q_t^n) a_{i,t}. \end{aligned}$$

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- Banker and fund problem is similar
 - ▶ No terminal utility, and no penalty on holding capital $\rightarrow$ aggregates into rep-agent.
 - ▶ Quantitative analysis: **Regulatory parameters** to discipline capital holdings.
- Equilibrium is standard, admits recursive characterization.

Methodology

Recursive Equilibrium: Master Equation

- Individual state = a_i , Aggregate states = (z, K, g)
- Household i chooses $(c_i, \theta_i^k, \theta_i^n, \theta_i^d, \iota_i)$ to solve value function $V(a, z, K, g)$ that satisfies

$$\begin{aligned} (\rho + \lambda_h)V = \max_{c_i, \theta_i, \iota_i} & \left\{ u(c_i) - \psi(\theta_i^k, a) + \lambda_h (\mathcal{U}(\mathcal{C}_i) - V) \right. \\ & + \partial_a V \mu_a + \partial_z V \mu_z + \partial_K V \mu_K + 0.5 (\partial_{a^2}^2 V \sigma_a^T \sigma_a + \partial_{z^2}^2 V \sigma_z^2) \\ & + \partial_{az}^2 V \sigma_{a,z} \sigma_z + \sigma_\lambda \sigma_V \\ & \left. \right\} \end{aligned}$$

Recursive Equilibrium: Master Equation

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$$\begin{aligned}(\rho + \lambda_h)V = & \max_{c_i, \theta_i, \iota_i} \left\{ u(c_i) - \psi(\theta_i^k, a) + \lambda_h (\mathcal{U}(\mathcal{C}_i) - V) \right. \\ & + \partial_a V \mu_a + \partial_z V \mu_z + \partial_K V \mu_K + 0.5 (\partial_{a^2}^2 V \sigma_a^T \sigma_a + \partial_{z^2}^2 V \sigma_z^2) \\ & + \partial_{az}^2 V \sigma_{a,z} \sigma_z + \sigma_\lambda \sigma_V \\ & \left. + \int_{\mathcal{A}} \partial_g V \mu_g + \int_{\mathcal{A}} \partial_{gz}^2 V \sigma_{g,z} \sigma_z + 0.5 \int_{\mathcal{A}} \int_{\mathcal{A}} \partial_{g,g}^2 V \sigma_g^T \sigma_{g'} \right\}\end{aligned}$$

Recursive Equilibrium: Hybrid approach

- Individual state = a_i , Aggregate states = $(z, K, \mathbf{g})$
- Household i chooses $(c_i, \theta_i^k, \theta_i^n, \theta_i^d, \iota_i)$ to solve value function $V(a, z, K, \mathbf{g})$ that satisfies

$$\begin{aligned}(\rho + \lambda_h)V = \max_{c_i, \theta_i, \iota_i} & \left\{ u(c_i) - \psi(\theta_i^k, a) + \lambda_h (\mathcal{U}(\mathcal{C}_i) - V) \right. \\ & + \partial_a V \mu_a + \partial_z V \mu_z + \partial_K V \mu_K + 0.5 (\partial_{a^2}^2 V \sigma_a^T \sigma_a + \partial_{z^2}^2 V \sigma_z^2) \\ & + \partial_{az}^2 V \sigma_{a,z} \sigma_z + \sigma_\lambda \sigma_V \\ & \left. + \sum_{\mathcal{A}} \partial_a V \mu_{a_j} + \sum_{\mathcal{A}} \partial_{az}^2 V \sigma_{a_j,z} \sigma_z + 0.5 \sum_{\mathcal{A}} \sum_{\mathcal{A}} \partial_{a,a}^2 V \sigma_a^T \sigma_{a'} \right\}\end{aligned}$$

Recursive Equilibrium: Hybrid approach

- Individual state = a_i , Aggregate states = $(z, K, \mathbf{g})$
- Household i chooses $(c_i, \theta_i^k, \theta_i^n, \theta_i^d, l_i)$ to solve value function $V(a, z, K, \mathbf{g})$ that satisfies

$$\begin{aligned}
 (\rho + \lambda_h)V = \max_{c_i, \theta_i, l_i} & \left\{ u(c_i) - \psi(\theta_i^k, a) + \lambda_h (\mathcal{U}(\mathcal{C}_i) - V) \right. \\
 & + \partial_a V \mu_a + \partial_z V \mu_z + \partial_K V \mu_K + 0.5 (\partial_{a^2}^2 V \sigma_a^T \sigma_a + \partial_{z^2}^2 V \sigma_z^2) \\
 & + \partial_{az}^2 V \sigma_{a,z} \sigma_z + \sigma_\lambda \sigma_V \\
 & \left. + \sum_{\mathcal{A}} \partial_a V \mu_{a_j} + \sum_{\mathcal{A}} \partial_{az}^2 V \sigma_{a_j,z} \sigma_z + 0.5 \sum_{\mathcal{A}} \sum_{\mathcal{A}} \partial_{a,a}^2 V \sigma_a^T \sigma_{a'} \right\}
 \end{aligned}$$

- Kolmogorov Forward Equation (KFE) for wealth evolution:

$$\begin{aligned}
 dg_h(a) = & \lambda_h \check{\phi}_h(a) g_h(a) da - \lambda_h g_h(a) & (\text{birth/death}) \\
 & - \partial_a [\mu_a a g_h(a)] + \frac{1}{2} \partial_a^2 [\sigma_a^2 a^2 g_h(a)] dt \\
 & - \partial_a [\sigma_a a g_h(a)] dW_z
 \end{aligned}$$

Methodology: Blocks

- Optimization block: Euler equation (in SDF ξ_h) and FOCs.

$$\begin{aligned}\rho &= r_d + \mu_{\xi,h}; & \xi_h &= u'(c_h); & \mu_{\xi,h} &= \text{drift of } \xi_h \\ r^l - r^d &= -\sigma_{\xi,h}\sigma_q - \text{Wedge}^l; & l &\in \{k, n, m\}\end{aligned}$$

- Distribution block:

$$\begin{aligned}dg_h(a) &= \lambda_h \check{\varphi}_h(a) g_h(a) da - \lambda_h g_h(a) && \text{(birth/death)} \\ &- \partial_a [\mu_a a g_h(a)] + \frac{1}{2} \partial_a^2 [\sigma_a^2 a^2 g_h(a)] dt \\ &- \partial_a [\sigma_a a g_h(a)] dW_z\end{aligned}$$

- Market clearing: Implicit price processes (q^k, q^n, q^m) consistent with Ito's lemma.

Methodology: "High level" Algorithm

1. Solving for equilibrium functions:

- ▶ Take finite dimensional approximation to agent continuum to get finite state space $\hat{\mathbf{S}}$.
 $\hat{\mathbf{S}} = \{z, K, \eta_1, \eta_2, \dots, \eta_I, \eta_f, \eta_b\}$, where $\eta_i = a_i/A$ is the wealth share of agent i .
- ▶ Represent equilibrium functions as DNN with states $\hat{\mathbf{S}}$ as inputs.
- ▶ Train DNN parameters to minimize loss in equilibrium conditions at random states.
Use vectorized automatic differentiation to compute gradients.

2. Simulation: use eqm functions to approximate KFE operator on a wealth grid.

Approach is easy to describe, tricky to implement.

"Art" of deep learning is to rewrite the problem to help the neural network learn.

Methodology: Loss Functions

- Define **consumption-to-wealth ratios** ($\omega := c/a$, $\Omega := \mathcal{C}/a$) and **p/f share** $\theta_j^l := q^l/a_j$.
- Represent ω , Ω , θ_h^l , q^l , μ_{q^k} , and σ_q by neural networks with parameters Θ .
- At state X , define the NN representation losses using

$$\hat{\Xi} = u'(\hat{\omega}(S)\eta\hat{q}^k(S)K).$$

$\mathcal{L}_{\omega} = r^d - \tau_h - \rho_h - \lambda_h + \mu_{\Xi} + D_{\theta}\psi$	Euler eq.
$\mathcal{L}_{\Omega} = \hat{\Omega} - \eta A \mathcal{W}(\theta_h^k, \theta_h^n)$	Retirement cons.
$\mathcal{L}_{\theta_h^l} = r^l - r^d + \sigma_{\xi_i} \cdot \sigma_{q^l} + \text{Wedge}$	FOC conditions
$\mathcal{L}_{\mu_{q^k}} = (D_S q^l)^T \mu_S + \frac{1}{2} \text{tr}(\sigma_S^T \sigma_x D_S^2 q^l)$	Consistency
$\mathcal{L}_{\sigma} = \hat{\sigma}_q - \sigma_S^T D_S q^l$	Consistency.

Finite Population: Why Are 10-20 Agents Sufficient?

Simulation-based finite-agent methods in discrete time often need a large number of agents to be accurate. Why not here?

1. We sample individual states (x) and distribution states ($\hat{\eta}$) separately when generating training sets
 - ▶ Without separation: need large N to have sufficiently many x in high-curvature regions (e.g., close to borrowing constraint) to learn value function well.
 - ▶ With separation: can independently control where we learn V in the x -dimension and how we represent g .
 - ▶ Ultimately, 10-20 agents is sufficient for approximating key moments of distribution.
2. We can eliminate/reduce finite agent noise
 - ▶ During training, we can eliminate the finite agent noise analytically in the “KFE”.
 - ▶ During simulation: approximate KFE operator instead of simulating finite pop. economy.